Instruction For Reproducing Section 4.2 Under PxG

Covariate-Assisted Bayesian Graph Learning for Heterogeneous Data

Description: This R Markdown file provides the workflow for reproducing simulation results in §4.2 under the PxG model.

1. Generating Simulation Data

```
rm(list = ls())

## reading functions for the simulation study
## "PxG-GW.R" contains all functions for Gaussian-G-Wishart model (section 3.2.1)
source("PxG-GW.R")

## generating simulation data from the 5-dimensional GGM in section 4.2
## first, generating covariate x
set.seed(2020)
x = runif(300, -0.8, 0.8)

## true precision matrix as a linear function of x
Omega = function(x){
  G0 = diag(5) - diag(5)
  G0[1,2] = G0[2,3] = G0[3,4] = G0[4,5] = 1
  G0 = G0 + t(G0)

  Omega = x*G0
  diag(Omega) = 1.4
  return(Omega)
}

## true graph
G0 = diag(5) - diag(5)
G0[1,2] = G0[2,3] = G0[3,4] = G0[4,5] = 1
G0 = G0 + t(G0)
diag(G0) = 1

## generating response y according to the true precision matrix
y = matrix(0, 300, 5)
for(i in 1:300){
  temp = solve(Omega(x[i]))
  y[i,] = t(t(chol(temp))%*%rnorm(5,0,1))
}
X = matrix(x, 300, 1); Y = y
```

2. Fitting Simulation Data With PxG

```
## fitting simulation data with the proposed PxG model
set.seed(2020)
A = PxG_GW(Y = Y, X = X, b = 3, D = diag(5), alphaG = 0.2,
          mu0 = as.vector(rep(0,1)), sigsq0 = 1, b1 = 10, b2 = 0.1, M = 1, N = 15,
          burnin = 1000, nmcmc = 2000, z0 = sample(1:15, 300, TRUE, rep(1/15, 15)), B = 3)
```

3. Preparation For Plots

```
## generating true partial correlation sequences for plot purposes
xseq = seq(-0.8, 0.8, by = 0.01)
Gseq = Oseq = array(0, c(5,5,length(xseq)))
for(i in 1:length(xseq)){
  Oseq[,,i] = Omega(xseq[i])
}

## true partial correlation sequences
w12 = Oseq[1,2,]/1.4
w23 = Oseq[2,3,]/1.4
w34 = Oseq[3,4,]/1.4
w45 = Oseq[4,5,]/1.4

## loading libraries for plot
library(ggplot2)
library(ggpubr)
## true partial correlation for each sample of covariate x
truerho = x/1.4
```

4. Generating Figure 5 and Display MSE

```
a = rep(0, length(x))
for(i in 1:length(x)) a[i] = A$Osum[1,2,i]/sqrt(A$Osum[1,1,i]*A$Osum[2,2,i])
## MSE for edge 1-2
MSE12 = mean((a-truerho)^2); MSE12 # 0.0077
```

```
## [1] 0.007740904
```

```
temp1 = data.frame(x1=xseq, y1=w12)
post = lowess(x, a, f=0.4)
temp2 = data.frame(x2=post$x, y2=post$y)
p1 = ggplot() +
  geom_line(temp1, mapping=aes(x=x1,y=y1,
                               color = "true partial correlation"), size = 0.6) +
  geom_line(temp2, mapping=aes(x=x2,y=y2,
                               color = "PxG posterior estimate"), size = 0.6) +
  scale_colour_manual(values = c("true partial correlation" = "black",
                                 "PxG posterior estimate" = "red")) +
  xlab("x") + ylab(expression(rho[12])) + labs(color = "") +
  theme(axis.text=element_text(size=12),
        axis.title.x=element_text(size=16),
        axis.title.y=element_text(size=16,face="bold"),
        legend.text = element_text(size=14)) +
  annotate("text", x = -0.35, y = 0.45, label = "MSE 0.0077")
```

```
a = rep(0, length(x))
for(i in 1:length(x)) a[i] = A$Osum[2,3,i]/sqrt(A$Osum[2,2,i]*A$Osum[3,3,i])
## MSE for edge 2-3
MSE23 = mean((a-truerho)^2); MSE23 # 0.0078
```

```
## [1] 0.007751489
```

```
temp1 = data.frame(x1=xseq, y1=w23)
post = lowess(x, a, f=0.4)
temp2 = data.frame(x2=post$x, y2=post$y)
p2 = ggplot() +
  geom_line(temp1, mapping=aes(x=x1,y=y1,
                               color = "true partial correlation"), size = 0.6) +
  geom_line(temp2, mapping=aes(x=x2,y=y2,
                               color = "PxG posterior estimate"), size = 0.6) +
  scale_colour_manual(values = c("true partial correlation" = "black",
                                 "PxG posterior estimate" = "red")) +
  xlab("x") + ylab(expression(rho[23])) + labs(color = "") +
  theme(axis.text=element_text(size=12),
        axis.title.x=element_text(size=16),
        axis.title.y=element_text(size=16,face="bold"),
        legend.text = element_text(size=14)) +
  annotate("text", x = -0.35, y = 0.45, label = "MSE 0.0078")

a = rep(0, length(x))
for(i in 1:length(x)) a[i] = A$Osum[3,4,i]/sqrt(A$Osum[4,4,i]*A$Osum[3,3,i])
## MSE for edge 3-4
MSE34 = mean((a-truerho)^2); MSE34 # 0.0078
```

```
## [1] 0.007795394
```

```
temp1 = data.frame(x1=xseq, y1=w34)
post = lowess(x, a, f=0.4)
temp2 = data.frame(x2=post$x, y2=post$y)
p3 = ggplot() +
  geom_line(temp1, mapping=aes(x=x1,y=y1,
                               color = "true partial correlation"), size = 0.6) +
  geom_line(temp2, mapping=aes(x=x2,y=y2,
                               color = "PxG posterior estimate"), size = 0.6) +
  scale_colour_manual(values = c("true partial correlation" = "black",
                                 "PxG posterior estimate" = "red")) +
  xlab("x") + ylab(expression(rho[34])) + labs(color = "") +
  theme(axis.text=element_text(size=12),
        axis.title.x=element_text(size=16),
        axis.title.y=element_text(size=16,face="bold"),
        legend.text = element_text(size=14)) +
  annotate("text", x = -0.35, y = 0.45, label = "MSE 0.0078")

a = rep(0, length(x))
for(i in 1:length(x)) a[i] = A$Osum[4,5,i]/sqrt(A$Osum[4,4,i]*A$Osum[5,5,i])
## MSE for edge 4-5
MSE45 = mean((a-truerho)^2); MSE45 # 0.0125
```

```
## [1] 0.01249844
```

```
temp1 = data.frame(x1=xseq, y1=w45)
post = lowess(x, a, f=0.4)
temp2 = data.frame(x2=post$x, y2=post$y)
p4 = ggplot() +
  geom_line(temp1, mapping=aes(x=x1,y=y1,
                               color = "true partial correlation"), size = 0.6) +
  geom_line(temp2, mapping=aes(x=x2,y=y2,
                               color = "PxG posterior estimate"), size = 0.6) +
  scale_colour_manual(values = c("true partial correlation" = "black",
                                 "PxG posterior estimate" = "red")) +
  xlab("x") + ylab(expression(rho[45])) + labs(color = "") +
  theme(axis.text=element_text(size=12),
        axis.title.x=element_text(size=16),
        axis.title.y=element_text(size=16,face="bold"),
        legend.text = element_text(size=14)) +
  annotate("text", x = -0.35, y = 0.45, label = "MSE 0.0125")

p0 = ggarrange(p1, p2, p3, p4, ncol = 4, nrow = 1, common.legend = TRUE); p0
```

